

The Global Fertility Dividend: Smaller Families and the Fall in Absolute Poverty

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Abstract

Global poverty is measured using household resources per person. When family size falls, resources per person rise mechanically, reducing poverty and causing demographic change to be recorded as income growth. I decompose the contribution of this “fertility dividend” under the assumption that the rank-ordering of co-resident children per mother is preserved across adjacent surveys using data on 22 countries that together held 81% of the world’s \$3.00-per-day poor in 1990. Smaller families account for 14.0% of the all-person decline at \$3.00 per day, 18.8% of the child-poverty decline, and 16.7% of the decline among people in households with children. The mechanism is robust to alternative rank assignments and equivalence scales.

Keywords: Poverty; fertility; family size; equivalence scales; demographic change

JEL Classification: I32, J13, O15

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1 Introduction

Global absolute poverty fell dramatically over the past forty years (World Bank, 2026). A household is defined to be poor when its income or consumption per member falls below a poverty line. The standard accounting of poverty change divides the decline into growth in household resources and changes in their distribution (Datt and Ravallion, 1992), and applications of it find that growth in mean incomes explains most of the fall (Kraay, 2006; Bergstrom, 2022). This paper asks how much of the measured improvement instead reflects demographic change in the denominator. As families had fewer children, household resources were spread across fewer people, reducing measured poverty even with resources held fixed. This direct resource-dilution channel is the microeconomic analog of a fertility dividend.

This dividend is largely absent from poverty accounting, including the World Bank–UNICEF estimates of child poverty (Salmeron-Gomez et al., 2023; Lara Ibarra et al., 2025). When family size falls and resources are held constant, income per person rises, and demographic change may be recorded as income growth rather than as an independent driver of poverty decline. The resulting “dividend” echoes the macroeconomic demographic dividend, but within families: fewer children share a given amount of household resources, raising resources relative to the poverty threshold. In the context of the global decline in fertility, this implies that the contribution of growth in household resources to reductions in poverty is overstated and the contribution of demographic change is understated. One reason this distinction matters is because income growth can be a persistent source of poverty reduction, whereas the fertility transition happens once. Household resources per child fall mechanically with family size, the budget-constraint premise underlying the quantity–quality tradeoff of Becker (1960) and Becker and Lewis (1973). Unless household resources rise enough to offset the larger denominator, the household moves closer to poverty.

The key identification challenge is the counterfactual distribution of children across families: which mothers, at which points in the resource distribution, would have had fewer children? I construct these counterfactuals by assuming that mothers’ percentile rank in

the fertility distribution is preserved between surveys. Within specified cells, each mother in the base sample is reassigned the number of children at the same percentile of the target distribution holding all other household characteristics fixed. This rank-preserving transport (Novosad, Rafkin and Asher, 2022) changes family size and, with it, resources per person and population weights, while other characteristics, including household resources and other residents, remain unchanged. I then recompute poverty under the counterfactual distribution. The design follows the United States analysis (Looney, 2026).

The approach relies on a rank-preservation assumption: within cells, mothers retain their ordinal position in the co-resident child distribution as that distribution shifts across surveys. This assumption is weaker than assuming fertility is independent of resources. High-fertility mothers may be poorer, so long as their relative fertility ranks are stable as the distribution changes. Because the number of children is discrete, ranks within child counts are not observed. In the main specification, I form ranks within rural and urban cells and use random assignment to break ties. I also report results from a range of alternative tie-breaking rules, including a copula calibrated to the observed welfare–fertility association, ordering by mother age, and monotone selection by welfare rank. These alternative rules move the estimate within a narrow band.

I apply the transport to 79 household survey waves in 22 countries that together held 81% of the world’s \$3.00-per-day poor in 1990, chaining adjacent survey pairs from 1990 to 2019. The data are national household expenditure and income surveys—the same surveys underlying the World Bank’s official poverty estimates. I supplement these surveys with census microdata from IPUMS International, which extend family-size information to time periods without resource data.

Smaller families account for 14.0% of the decline in all-person \$3.00-per-day poverty, an even larger share (18.8%) of the decline in child poverty, and 16.7% of the decline among people in households with children—a cumulative contribution of 160 million people. In countries where poverty rose, fertility decline often cushioned the increase. At higher poverty

thresholds, the all-person share is 11.0% at \$4.20 and 9.9% at \$8.30.

The results isolate a narrow, mechanical resource-dilution effect of fertility on poverty, not the total effect of fertility change, which includes behavioral channels that likely also affected poverty. In particular, this analysis excludes the effects of delayed childbearing and smaller families on parental labor supply, educational attainment, investments in children, or public budgets. To the extent those channels change observed household resources, this analysis attributes them to income or consumption growth.

The paper makes several contributions. First, it demonstrates that the decline in fertility and family size was a substantial contributor to the fall in global poverty. Second, it uses a rank-preserving transport method to decompose changes in poverty. The data cannot reveal which mothers would have had fewer children, so the paper reports ranges of estimates over a set of economically restricted rank assignments, in the spirit of partial identification (Manski, 2003; Tamer, 2010). Third, it applies these methods across 22 countries and 79 survey waves. Finally, it extends the analysis to a longer horizon using IPUMS International censuses (Ruggles et al., 2024). Because censuses lack comparable household welfare, these are one-directional implied effects at an observed survey’s living standards, not observed poverty decompositions.

1.1 Relationship to the literature

A macroeconomic demographic-dividend literature studies how age structure affects aggregate saving, labor supply, and growth (Bloom and Williamson, 1998; Ashraf, Weil and Wilde, 2013). International poverty applications similarly relate fertility, dependency ratios, and poverty in macro or grouped data (Eastwood and Lipton, 1999; Cruz and Ahmed, 2018). Wietzke (2020) asks a similar question in cross-country regressions of poverty on lagged fertility. The effects of fertility and demographics in this literature therefore reflect a combination of factors including the number of workers, age composition, household formation, human-capital investment, labor supply effects, and general-equilibrium growth. I ask a

more specific question: holding observed household resources fixed, how much does measured poverty change simply because of changes in the number of co-resident children sharing those resources?

Decompositions of changes in poverty over time conventionally divide a change into growth and redistribution components (the Datt–Ravallion decomposition; Datt and Ravallion, 1992; Ravallion, 2016). Applications to large country panels attribute most of the world’s poverty reduction to growth in the mean rather than to redistribution (Kraay, 2006; Bergstrom, 2022). In these studies, welfare is already expressed per person or per equivalent adult, so any effect of fertility or demographic change is already incorporated as growth. World Bank applications add a demographic term by substituting the adult share or household-size ratio between surveys (Azevedo et al., 2013; Inchauste et al., 2014). However, that approach confounds fertility with changes in co-residence or aging, and it does not preserve the joint distribution of family size and resources near the poverty line. Recent World Bank–UNICEF analyses document large declines in monetary child poverty measured with household income or consumption per capita (Salmeron-Gomez et al., 2023; Lara Ibarra et al., 2025), but they do not separate changes in household resources from changes in the number of people sharing them.

A large literature asks how household size and economies of scale should enter poverty measurement. Larger households may share housing, durable goods, fuel, and other public goods, while children may require fewer resources than adults (Lanjouw and Ravallion, 1995; Deaton and Paxson, 1998; Jolliffe and Tetteh-Baah, 2024). That literature mainly concerns how to define poverty levels—who is poor under a per-capita, adult-equivalent, or scale-adjusted definition—while the question here asks why household resources change relative to those thresholds. Assessing sensitivity to alternative scales therefore also quantifies how much of the historical poverty change depends on the normative weight assigned to an additional child under these alternative scales.

This analysis also relates to the literature on differential fertility. Poorer and less-

educated women often have more children, and the gradient changes during development (Vogl, 2016). This selection is why the joint distribution of fertility and income matters. A shift of one child concentrated among households near the poverty line may have a different effect from the same marginal shift among richer households. The rank-preserving transport retains the observed relationship between income and fertility.

2 Data and measurement

The analysis draws on microdata from national household surveys, harmonized to match the World Bank’s Poverty and Inequality Platform (PIP; World Bank, 2026). From each survey I take the household roster, survey weights, urban or rural residence, and the income or consumption resource measure. From PIP I take the official country-year headcount, reporting population, poverty line, and welfare concept. The poverty lines are the World Bank’s international lines at 2021 purchasing power parities (Foster et al., 2025). I place the poverty line at the percentile of each survey’s own welfare distribution given by that country-year’s official PIP headcount, matched on the income-or-consumption concept PIP uses. This rescales the country’s resource measure by a single constant so that its headcount matches PIP. Because the rescaling is monotone, it changes consumption units without changing ranks or the identity of the poor. The anchored headcounts reproduce the official PIP headcounts to within 0.123 percentage points in every wave. In total, 79 survey waves form 57 adjacent pairs in 22 countries.

I focus on the years 1990–2019: 1990 is a common starting year for global poverty comparisons, and 2019 avoids the disruptions of COVID-19. Some adjacent survey pairs straddle 1990 or 2019. For these pairs, I prorate the observed poverty change and the fertility effect linearly according to the fraction of elapsed years inside the window. The appendix reports the sensitivity of the pooled share to this proration rule and to mother-link quality. It is small.

The primary fertility measure is the number of co-resident children ages 0–17 linked to each co-resident mother. I assign children to a plausible co-resident mother using a common probabilistic hierarchy. Where surveys record mother line numbers, the actual link supports the probabilistic rule. Children who cannot be linked are held fixed in the counterfactual (and thus treated the same as other household members). The transported quantity is therefore a measure of the co-resident family-size stock rather than the total fertility rate or completed fertility: changes in child survival, migration, age composition, and co-residence are not separately identified. Moreover, the role of the extensive margin between zero and one co-resident child lies outside the analysis.

I use all countries with publicly available survey data, including household composition and a measure of income or consumption. I exclude surveys where fewer than 80 percent of household children can be linked to a co-resident mother. This rule excludes a few early files whose household rosters do not include usable mother pointers (Peru before 2004, Kenya 1992). With this restriction, the analysis uses 22 countries with usable income or consumption, household composition, and mother-level roster information.¹

I use countries represented only in IPUMS for describing long-horizon demographic changes, but not in the decomposition of poverty.

The sample thus covers most of the relevant population, with two caveats. First, the Chinese data ends in 2002 and there is no comparable welfare microdata after that year, and no post-2000 IPUMS sample. Most of China’s poverty decline occurred after 2002 (World Bank, 2026), so the largest national episode in the global series is only partly covered. The effect of the omission on the pooled share is ambiguous, because the omitted years drop out of the numerator and the denominator alike. Second, India’s NSS and HCES rosters do not include mother line numbers, so every Indian pair relies on the probabilistic mother assignment.

¹The Philippines has a long survey series with under-15 counts but no mother link in the relevant files. Household-level results from the Philippines are reported in the appendix but are not included in the mother-level analysis.

Family size fell substantially over the sample windows. Weighting countries by survey population, the mean number of linked children per mother fell from 2.34 in each country’s first analysis wave to 1.92 in its last; Table A1 reports each country’s first- and last-wave means.

Figure 1 documents the economic selection underlying the design. In every sample country and wave, mothers in the poorest welfare quintile have more co-resident children than mothers in the richest quintile. The gradient generally compresses as fertility falls but never reverses.

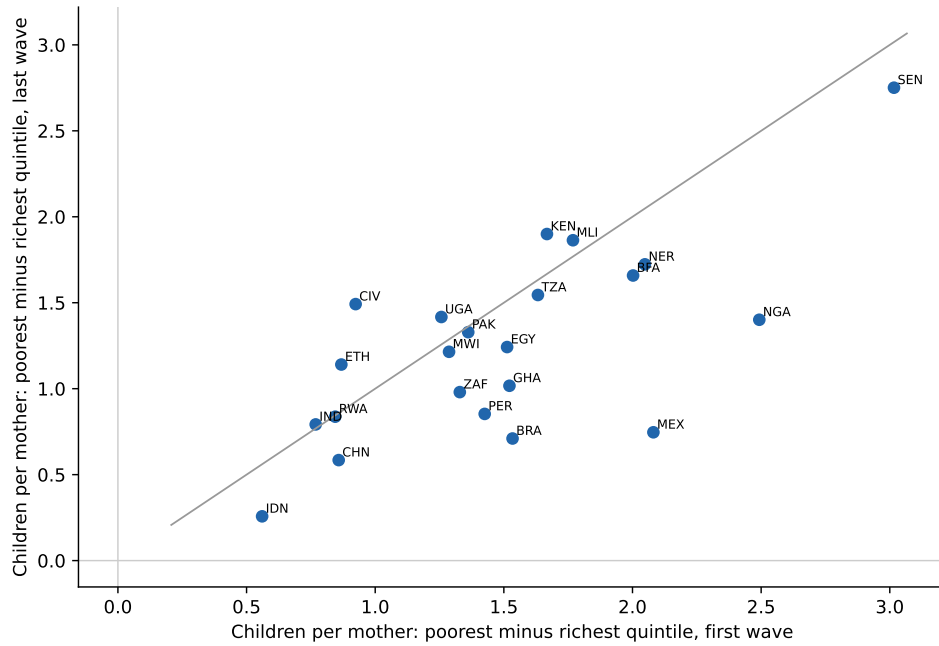


Figure 1: The welfare–fertility gradient persists as family size falls

Notes: Each point is a country. Axes show the survey-weighted difference in linked children per mother between the poorest and richest household-welfare quintiles in the first and last usable survey waves. The 45-degree line denotes an unchanged gradient.

3 Empirical method

Let household h have fixed resources R_h , fixed nonchildren A_h , and linked mothers m with child counts K_m , so that $K_h = \sum_{m \in h} K_m$. Under the official per-capita measure, welfare is

$$y_h = \frac{R_h}{A_h + K_h}.$$

Within urban/rural cell x , let $F_x^t(k)$ be the weighted distribution of children per linked mother in survey t . A mother at weighted rank p in the source distribution receives

$$K_m^{cf} = \min\{k : F_x^{t_1}(k) \geq p\}.$$

Counterfactual mother counts are summed within households, unlinked children and all nonchildren are held fixed, and poverty is recomputed from $R_h/(A_h + K_h^{cf})$. For child poverty, the household's poverty status is weighted by its counterfactual number of children. The transported children-per-mother distribution reproduces its target marginal in every pair and direction to within a total-variation distance of 0.38%.

The central counterfactual-assignment problem is how to attach the target family-size distribution to the source population of families and resources. The aggregate decline in family size (and its distribution) is observed, but it is not clear to which households to assign the changed fertility distribution. The identifying assumption in this paper is conditional rank preservation. Let u_m denote a mother's rank in her cell's resource distribution, v_m her latent fertility rank, and C_x^t their joint distribution within urban/rural cell x in survey t . The identifying condition is stability of that dependence structure across adjacent surveys:

$$C_x^{t_0}(u, v) = C_x^{t_1}(u, v).$$

In plain terms, the assumption is that mothers who tend to have more children than their peers in one era also tend to have more children in another. The condition makes no specific

assumption about how fertility is correlated with resources or observed characteristics, only that the relationship between fertility rank and other characteristics is preserved between adjacent surveys. Because child counts are discrete, the latent fertility rank is not defined within those integer counts where assignment is genuinely unidentified. The primary analysis breaks these ties with random assignment and I assess the robustness to alternative tie-breaking rules.

For each adjacent pair and each population $j \in \{\text{all}, \text{child}, \text{with-child}\}$, let D_F^j be the poverty reduction from imposing the later child distribution on the earlier survey, and let D_R^j be the poverty increase from imposing the earlier distribution on the later survey. Both are signed so that a positive value is a poverty reduction attributable to smaller families. The pair effect is

$$D^j = \frac{1}{2}(D_F^j + D_R^j).$$

The forward calculation measures demographic change holding fixed the earlier welfare distribution. The reverse calculation measures it at the later distribution. For each adjacent pair I use the average of the forward and reverse transports, then chain consecutive pair effects to produce each country's cumulative effect. I construct the pooled share from individual counts. For country-pair q , let N_q^j be the national population of group j , averaged across the pair's two survey years, from PIP reporting populations scaled by the group's survey share; let s_q be the exposure fraction, D_q^j the transport effect, and ΔH_q^j the observed change. The pooled statistic is

$$S^j = \frac{\sum_q s_q N_q^j D_q^j}{\sum_q s_q N_q^j \Delta H_q^j}.$$

This formula uses all compatible countries, including those with negative ΔH_q^j . The numerator is a cumulative sum of period-specific headcount contributions.

Alternative equivalence scales replace $A + K$ consistently in both the actual and counterfactual poverty measures and each wave is reanchored to its PIP headcount. I report the square-root, Betson (1996), OECD-modified, and adults-plus-half-a-child scales. Because

these scales give an additional child less than unit need, they should produce smaller effects than the official per-capita measure.

4 Results

At \$3.00 per day, smaller families account for 14.0% of the pooled all-person decline, 18.8% of the child-poverty decline, and 16.7% of the decline among people in households with children (Table 1). The child and with-children estimates are larger only because the affected population receives greater weight. The cumulative all-person numerator is 160 million. The pooled share divides it by the net change across all countries, including the 6 countries where poverty rose. Restricting the pool to countries with observed declines gives 13.5%, and to the 11 countries with declines above 10 percentage points gives 13.3%.

Table 1: Family-size contribution on a fixed all-country sample

Poverty line	All persons	Children	With-child HH	Effect (M)	Decline (M)	Countries
\$3.00	14.0%	18.8%	16.7%	160	1,142	22
\$4.20	11.0%	16.3%	13.9%	131	1,186	22
\$8.30	9.9%	18.2%	14.7%	45	460	22

Notes: Ratios are summed population-count transport effects divided by summed observed population-count declines. Effect (M) and Decline (M) report the all-person numerator and denominator of that ratio in millions of people; the numerator is the cumulative sum of pair-specific headcount effects across the observed survey intervals, not a count of distinct individuals. All outcomes average forward and reverse transports for every adjacent pair and sum the same pair chain. No country is selected on the sign or size of its poverty change.

Source: Author's calculations from national household surveys anchored to the World Bank Poverty and Inequality Platform (PIP).

Fertility decline reduced poverty in most countries even when aggregate poverty rose, acting as partial insurance against crises or stagnant household resources. Figure 2 shows every country's change in percentage points. Gray bars are the observed all-person poverty decline, so bars to the left of zero identify countries where poverty rose. Colored symbols show the family-size effects.

In Mexico, Brazil, and Peru, family-size effects are at least as large at \$4.20 and \$8.30 as at \$3.00 (Figure 3). That pattern is intuitive: extreme poverty can already be rare, but

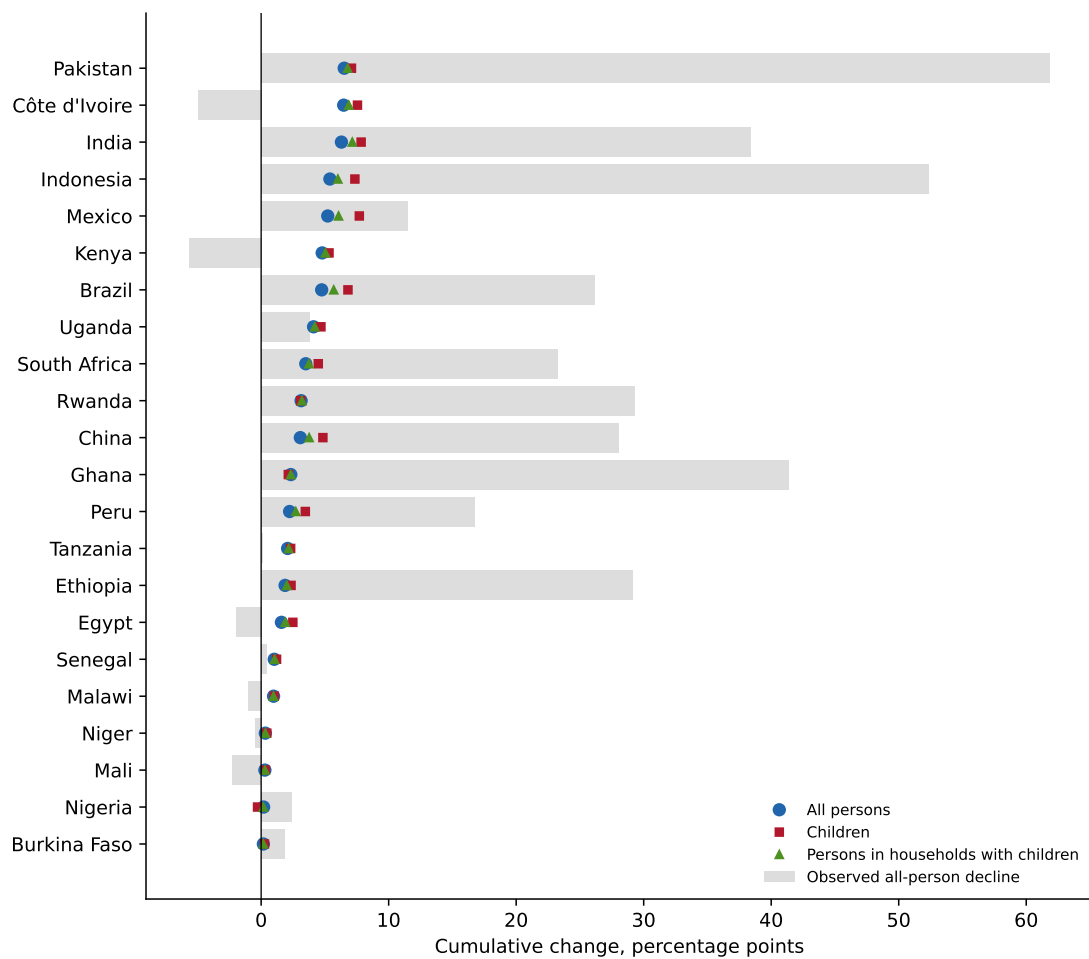


Figure 2: Falling family size reduced poverty even where overall poverty rose
Notes: Gray bars show cumulative observed all-person changes; positive values are poverty declines. Symbols show symmetric family-size effects for the same adjacent pairs. Countries are not selected on the sign or magnitude of the observed change.

if the density of households is high at alternative thresholds, changes in fertility can have sizable effects. In China and Indonesia, the higher-line effect is smaller over the observed window because rapid growth moved much of the welfare distribution past those thresholds. The pooled fixed-sample shares are 14.0%, 11.0%, and 9.9%, respectively.

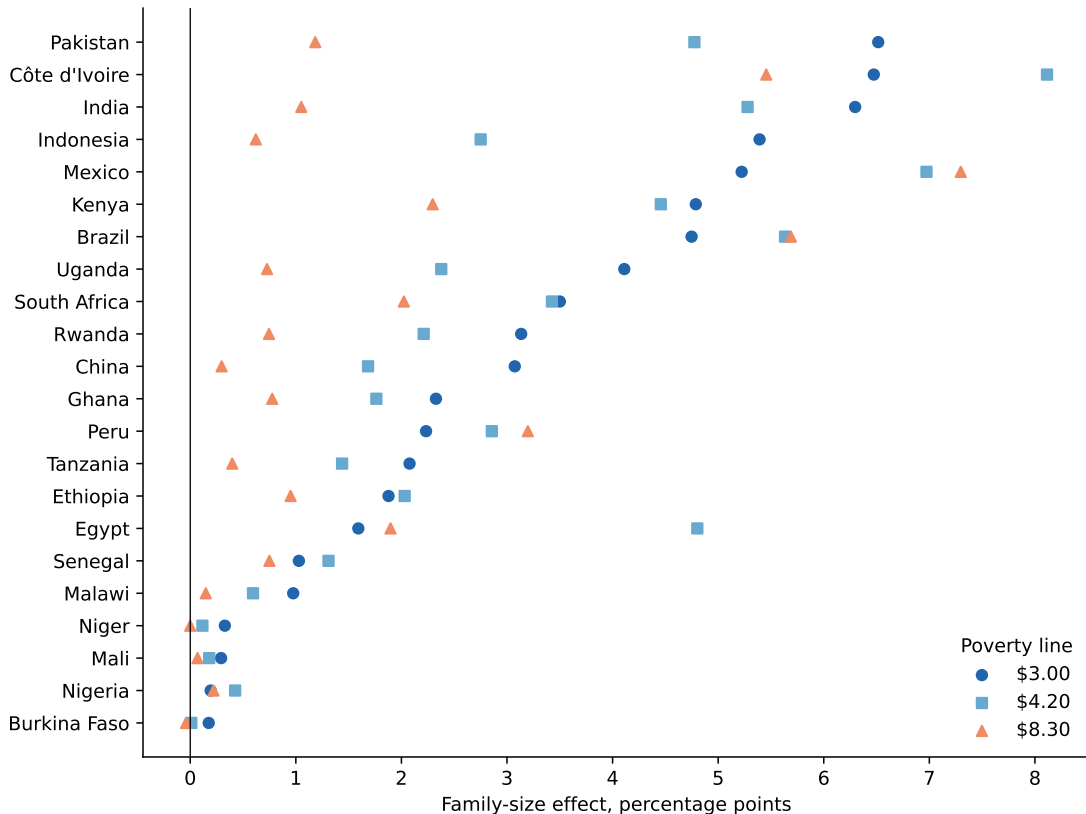


Figure 3: The poverty-line response differs across development paths
Notes: All-person percentage-point effects use the same countries, survey pairs, symmetric estimator, and 1990–2019 exposure at each line. Each wave is reanchored to PIP at the indicated line.

Standard growth–redistribution decompositions place this channel in the growth component because they compare welfare per person across surveys after controlling for household size.

4.1 Why the magnitude varies across countries

The family-size share is smallest where household resources grew fastest. In fast-growth episodes, the family-size effect can be sizable in percentage points but a modest fraction of

a large poverty decline (Figure 2). In crisis or low-growth episodes, the same demographic movement is often the principal poverty-reducing force.

Another reason the magnitude varies across countries is differences in the welfare–fertility gradient. Because large families are concentrated among poorer households in every sample country and wave (Figure 1), transporting the later, smaller-family distribution moves more households across a low poverty line.

Finally, the density of welfare near the poverty line determines how many households are shifted across the line because of fertility changes. A mechanical change in resources per person affects measured poverty only to the extent it moves the household and its residents across the line. If few households lie near a line, even a large average denominator change has little effect. Conversely, a smaller demographic change can produce a large poverty effect when the welfare distribution is dense around the line.

5 Identification, robustness, and long-horizon evidence

The estimates above are robust to how the unidentified ties within child counts are broken. Panel A of Figure 4 reports ranges over economically restricted rank assignments, following the logic of bounds over admissible couplings (Manski, 2003). Instead of totally unrestricted bounds, I focus on economically motivated tie breaking rules: random completion (the baseline); a Gaussian copula calibrated to the observed welfare–fertility association; ordering by mother age, which lets fertility timing structure the ranks; and monotone selection, under which poorer mothers have weakly higher latent fertility within a child-count mass point. Across the four assignment rules, the all-person share runs from 13.6% to 14.7%, the child share from 18.8% to 20.4%, and the with-children share from 16.4% to 17.6%.

The rank-preservation assumption itself can also be relaxed. In this test, each mother’s transport position is a mixture of her observed fertility rank and an independent random draw. The mixing weight λ runs from 0 to 1. At $\lambda = 1$, the target family-size marginal

is assigned at random within urban and rural cells and rank preservation plays no role. The pooled all-person share is 14.1% at $\lambda = 0$, 14.2% at $\lambda = 0.25$, 14.4% at $\lambda = 0.5$, and 14.9% under fully random assignment. (14.1% differs slightly from 14.0% because the within-tie completion draws are redrawn.) The estimate is stable. Rank preservation appears conservative because it keeps the remaining children concentrated among poor households, and random assignment yields a slightly larger family-size contribution.

The cell definition itself matters more than the within-cell assignment, and the urban/rural rule sits at the conservative end of the alternatives. Recomputing the pooled all-person share under 6 alternative partitions—a single national cell, mother-age bands, household adult counts, and crossings of these with the urban/rural divide—yields estimates from 13.9% to 16.6%. All but one of the alternatives produce a larger share, chiefly because partitions that pool urban and rural mothers let the transport attribute the rural–urban fertility differential, which is correlated with poverty, to the family-size margin. Placebo partitions that assign households to random cells reproduce the single-cell estimate (15.4% and 15.5% against 15.5%), so the spread across the informative partitions reflects their covariate content rather than the mechanics of splitting the sample.

Conditional household bootstrap intervals, which quantify sampling-related uncertainty, are 13.7%–14.3% for all persons, 18.4%–19.3% for children, and 16.3%–17.1% for people in households with children. The resampling uses common Poisson household weights for repeated appearances of a survey wave in adjacent pairs, while holding PIP anchors and the random transport completion fixed.

Relative to the official per-capita estimate of 14.0%, the square-root scale yields 7.1%, Betson 6.5%, OECD-modified 8.6%, and adults-plus-half-a-child 9.2%. The smaller estimates reflect the scales’ assignment of less additional need to a child because of economies of scale.

An alternative concern is that children may contribute to household resources. Allowing household resources to fall by 5 or 10 percent for each removed child lowers the estimate to 10.0% and 5.8%. Those scenarios exceed the measured direct contribution of children to

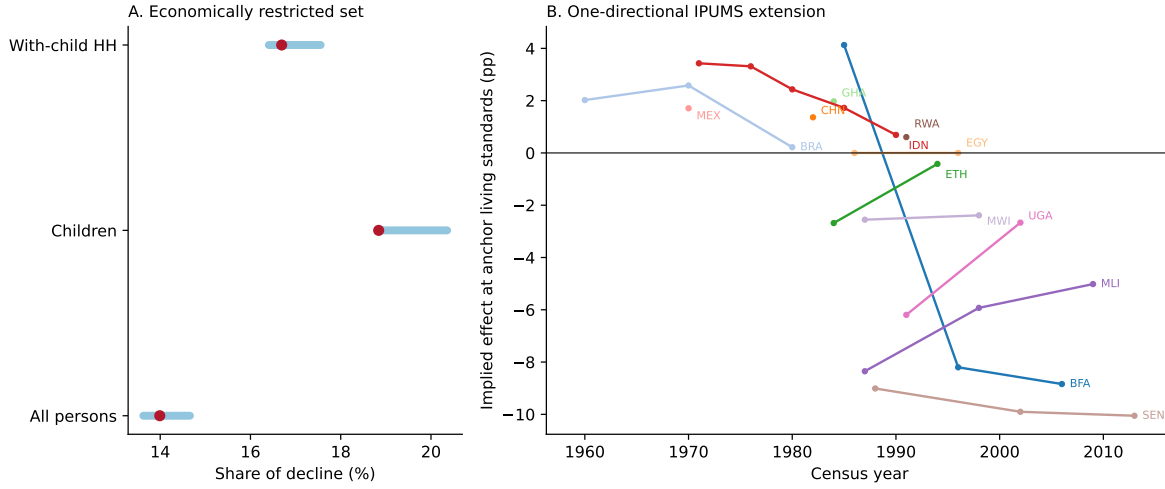


Figure 4: Economically restricted assignments and one-directional census extensions

Notes: Panel A shows the range over the four declared economically admissible within-tie completions; red dots are random-completion estimates. Panel B imposes IPUMS census fertility marginals on the nearest observed survey welfare distribution for every reliable country in every non-overlap census year. These implied effects are not observed poverty changes.

household resources: in the two sample surveys that record person-level income by source, all members under 18 combined receive only 3.5% of household income among Mexican households with children and 1.6% in Brazil—less than the 5 percent the milder scenario assumes for each removed child.

Panel B extends the exercise to census years that record family size but no income or consumption. I hold the nearest survey’s welfare distribution fixed, impose the census family-size distribution, and read off how much higher or lower poverty would have been at that survey’s living standards. There is no symmetric reverse transport, because the census year has no welfare data. A positive value means families were larger at the census than at the survey that anchors it, so family size fell over the intervening years and the survey window captures only part of the contribution. A negative value means co-resident families were still growing when the survey window opened.

Both patterns appear, and they separate countries by their stage in the demographic transition. In Brazil, China, Indonesia, and Mexico, where fertility had been falling for decades before the first usable survey, children per mother were higher at the census than

at the anchor—2.8 in Indonesia in 1971 against 2.2 in 1993—so the survey-window estimate understates the longer-horizon contribution. Burkina Faso, Ghana, and Rwanda carry the same sign. In Ethiopia, Mali, Malawi, Senegal, and Uganda the implied effect is negative because children per mother were still rising as of the earliest census, from 2.9 to 3.8 in Senegal between 1988 and 2018. Falling child mortality outpaced falling fertility, so the number of children sharing a household’s resources grew, and the family-size channel raised poverty before the survey window rather than lowering it. Egypt’s implied effect is zero because almost no Egyptian households sit near the \$3.00 line, not because its family size was unchanged.

The IPUMS exercise uses exact census mother pointers and covers pre-survey, within survey, and post-survey census years where available. A country’s censuses are used in Panel B only if they satisfy two conditions: (1) the census links children to mothers cleanly, and (2) wherever a census year falls within three years of a survey wave, the census and the survey agree on children per mother. The second condition can be tested only when a census falls that close to a survey. For countries without such an overlap, inclusion rests on the first condition alone. 14 countries meet the applicable conditions; Panel B plots the 13 of them that have at least one census year outside the survey window. Across 20 census–survey overlaps within two years, the median absolute children-per-mother difference is 0.10 and the median absolute implied poverty effect is 0.59 percentage points. The appendix compares nearest and distant survey anchors.

6 Conclusion

Declining family size was a substantial and overlooked contributor to the fall in global absolute poverty. Smaller families account for 14.0% of the all-person decline at \$3.00 per day, with larger contributions for children and people in households with children. Because conventional poverty accounting conditions on family or household size, this demographic

improvement is attributed to income or consumption growth.

The analysis holds household resources fixed, and thus excludes changes in parental labor supply, human capital, investments in children, or changes in public resources, which may also affect absolute poverty. Each country's fertility transition happens once, so the contribution measured here will not repeat. Where family size has already fallen, further reductions in measured poverty must come from household resources rather than from the denominator.

Data Availability Statement

Code, generated results, tests, and provenance records form a replication archive available from the author. Licensed household microdata is publicly available but cannot be redistributed. Public PIP and IPUMS inputs are documented by source and vintage.

References

- Ashraf, Quamrul H., David N. Weil, and Joshua Wilde. 2013. “The Effect of Fertility Reduction on Economic Growth.” *Population and Development Review*, 39(1): 97–130.
- Azevedo, João Pedro, Gabriela Inchauste, Sergio Olivieri, Jaime Saavedra, and Hernan Winkler. 2013. “Is Labor Income Responsible for Poverty Reduction? A Decomposition Approach.” World Bank Policy Research Working Paper 6414.
- Becker, Gary S. 1960. “An Economic Analysis of Fertility.” In *Demographic and Economic Change in Developed Countries*. 209–240. New York: Columbia University Press.
- Becker, Gary S., and H. Gregg Lewis. 1973. “On the Interaction between the Quantity and Quality of Children.” *Journal of Political Economy*, 81(2): S279–S288.
- Bergstrom, Katy. 2022. “The Role of Income Inequality for Poverty Reduction.” *World Bank Economic Review*, 36(3): 583–604.
- Betson, David M. 1996. “Is Everything Relative? The Role of Equivalence Scales in Poverty Measurement.” U.S. Census Bureau Poverty Measurement Working Paper.
- Bloom, David E., and Jeffrey G. Williamson. 1998. “Demographic Transitions and Economic Miracles in Emerging Asia.” *World Bank Economic Review*, 12(3): 419–455.
- Cruz, Marcio, and S. Amer Ahmed. 2018. “On the Impact of Demographic Change on Economic Growth and Poverty.” *World Development*, 105: 95–106.
- Datt, Gaurav, and Martin Ravallion. 1992. “Growth and Redistribution Components of Changes in Poverty Measures.” *Journal of Development Economics*, 38(2): 275–295.
- Deaton, Angus, and Christina Paxson. 1998. “Economies of Scale, Household Size, and the Demand for Food.” *Journal of Political Economy*, 106(5): 897–930.
- Eastwood, Robert, and Michael Lipton. 1999. “The Impact of Changes in Human Fertility on Poverty.” *Journal of Development Studies*, 36(1): 1–30.
- Foster, Elizabeth M., Dean Jolliffe, Gabriel Lara Ibarra, Christoph Lakner, and Samuel Kofi Tetteh-Baah. 2025. “Global Poverty Revisited Using 2021 PPPs and New Data on Consumption.” World Bank Policy Research Working Paper 11137.
- Inchauste, Gabriela, João Pedro Azevedo, Boniface Essama-Nssah, Sergio Olivieri, Trang Van Nguyen, Jaime Saavedra-Chanduvi, and Hernan Winkler. 2014. *Understanding Changes in Poverty*. Washington, DC: World Bank. Directions in Development.
- Jolliffe, Dean, and Samuel Kofi Tetteh-Baah. 2024. “Identifying the Poor: Accounting for Household Economies of Scale in Global Poverty Estimates.” *World Development*, 179: 106593.

- Kraay, Aart.** 2006. “When Is Growth Pro-Poor? Evidence from a Panel of Countries.” *Journal of Development Economics*, 80(1): 198–227.
- Lanjouw, Peter, and Martin Ravallion.** 1995. “Poverty and Household Size.” *Economic Journal*, 105(433): 1415–1434.
- Lara Ibarra, Gabriel, Daylan Alberto Salmeron Gomez, Solrun Engilbertsdottir, Carolina Diaz-Bonilla, Enrique Delamonica, and Jennifer Yablonski.** 2025. “Children in Monetary Poor Households: Global, Regional, and Select National Trends in the Progress against Child Poverty.” World Bank Policy Research Working Paper 11203.
- Looney, Adam.** 2026. “The Fertility Dividend: Family Size and the Decline in Child Poverty.” The Brookings Institution Working Paper.
- Manski, Charles F.** 2003. *Partial Identification of Probability Distributions*. New York:Springer.
- Novosad, Paul, Charlie Rafkin, and Sam Asher.** 2022. “Mortality Change among Less Educated Americans.” *American Economic Journal: Applied Economics*, 14(4): 1–34.
- Ravallion, Martin.** 2016. *The Economics of Poverty: History, Measurement, and Policy*. New York:Oxford University Press.
- Ruggles, Steven, Lara Cleveland, Rodrigo Lovaton, Sula Sarkar, Matthew Sobek, Derek Burk, Dan Ehrlich, Quinn Heimann, and Jane Lee.** 2024. “Integrated Public Use Microdata Series, International: Version 7.5 [data set].” *Minneapolis, MN: IPUMS*. <https://doi.org/10.18128/D020.V7.5>.
- Salmeron-Gomez, Daylan, Solrun Engilbertsdottir, Jose Antonio Cuesta Leiva, David Newhouse, and David Stewart.** 2023. “Global Trends in Child Monetary Poverty According to International Poverty Lines.” World Bank Policy Research Working Paper 10525.
- Tamer, Elie.** 2010. “Partial Identification in Econometrics.” *Annual Review of Economics*, 2(1): 167–195.
- Vogl, Tom S.** 2016. “Differential Fertility, Human Capital, and Development.” *Review of Economic Studies*, 83(1): 365–401.
- Wietzke, Frank-Borge.** 2020. “Poverty, Inequality, and Fertility: The Contribution of Demographic Change to Global Poverty Reduction.” *Population and Development Review*, 46(1): 65–99.
- World Bank.** 2026. “Poverty and Inequality Platform (version 20260324.2021_01_02.PROD) [data set].” <https://pip.worldbank.org>, accessed June 10, 2026.

A Country coverage and survey chains

Table A1: All compatible country chains at \$3.00 per day

Country	Window	Observed (pp)	Effect (pp)	Share	Exact/pairs	Children/mother
Brazil	1990–2019	26.1	4.7	18.2%	1/3	2.37 → 1.58
Burkina Faso	2018–2019	1.9	0.2	9.3%	0/1	3.26 → 3.27
China	1990–2002	28.0	3.1	11.0%	0/2	1.81 → 1.37
Côte d’Ivoire	1990–2019	-5.0	6.5	–	0/2	3.52 → 2.69
Egypt	1999–2019	-2.0	1.6	–	7/7	2.62 → 2.38
Ethiopia	1999–2019	29.2	1.9	6.4%	1/3	2.93 → 2.46
Ghana	1991–2016	41.3	2.3	5.6%	1/2	2.94 → 2.35
India	1990–2019	38.4	6.3	16.4%	0/4	2.60 → 1.96
Indonesia	1993–2014	52.3	5.4	10.3%	4/4	2.24 → 1.69
Kenya	1994–2015	-5.7	4.8	–	0/3	3.29 → 2.67
Malawi	2004–2019	-1.0	1.0	–	0/1	2.73 → 2.42
Mali	2018–2019	-2.3	0.3	–	0/1	3.47 → 3.45
Mexico	1990–2019	11.5	5.2	45.4%	1/4	2.80 → 1.72
Niger	2018–2019	-0.5	0.3	–	0/1	3.45 → 3.39
Nigeria	2010–2019	2.4	0.2	8.1%	1/2	3.15 → 2.82
Pakistan	1991–2019	61.9	6.5	10.5%	3/6	3.78 → 2.79
Peru	2004–2019	16.8	2.2	13.3%	0/3	2.23 → 1.81
Rwanda	2000–2019	29.3	3.1	10.7%	0/2	2.79 → 2.34
Senegal	2018–2019	0.4	1.0	245.1%	0/1	3.81 → 3.47
South Africa	1993–2019	23.3	3.5	15.1%	0/3	2.51 → 2.02
Tanzania	2011–2017	0.1	2.1	2115.3%	0/1	2.90 → 2.71
Uganda	2009–2019	3.8	4.1	108.6%	0/1	3.23 → 2.80

Notes: A share is shown as a dash when poverty rose or was unchanged because the ratio is not interpretable; every row remains in the pooled count calculation. Exact/pairs counts adjacent survey pairs in which both waves carry recorded mother line numbers, out of all pairs in the chain. Children/mother is the survey-weighted mean number of linked co-resident children per linked mother in the country’s first and last analysis waves.

Source: Author’s calculations from national household surveys anchored to the World Bank Poverty and Inequality Platform (PIP).

All compatible mother-level chains remain in the fixed sample (Table A1). Shares are omitted for country rows in which poverty rose because a positive poverty-reducing effect divided by an observed increase in poverty is not informative. Those observations remain in every population-count pool.

The 1990–2019 boundary rule prorates a pair only when it crosses an endpoint. Boundary-prorated pairs contribute 34.6% of the cumulative all-person count numerator, so the linear-accrual assumption matters for the cumulative count. The pooled share is insensitive to it: restricting the pool to the 39 pairs lying entirely inside 1990–2019 gives 13.7%, and removing proration entirely, which admits each boundary pair’s full effect, gives 14.7% with a cumulative count of 189 million. Concept breaks are retained in the headline only when they are part of PIP’s own country series; restricting the pool to pairs that hold the welfare concept fixed, which removes only China 1988–1995, gives 14.2%. Datt–Ravallion comparators are computed as diagnostics and are never used to replace the primary survey pairs.

Mother-link quality does not drive the estimate. Pairs in which both waves carry recorded

mother line numbers and pairs that rely on the probabilistic assignment give nearly identical pooled shares (14.1% and 14.0%). Dropping India, whose rosters carry no mother line numbers, gives 13.0%. Leave-one-country-out shares range from 13.0% (dropping India) to 15.3% (dropping China).

The child-poverty effect is almost entirely a within-household denominator effect rather than a change in the number of children counted. Children who remain in the household cross the line as measured resources per person rise; this component accounts for 18.1% of the 18.8% child-poverty share, while the composition component from the smaller counterfactual child population contributes 0.7%.

B Equivalence scales and resource responses

Table A2: Equivalence-scale and resource-offset specifications

Specification	All	Child	With-child HH
Per capita (official)	14.0%	18.8%	16.7%
Square root	7.1%	9.4%	8.5%
Betson	6.5%	9.1%	7.9%
OECD modified	8.6%	12.1%	10.4%
Adults + 0.5 child	9.2%	13.3%	11.1%
5 percent resource offset	10.0%	13.0%	11.7%
10 percent resource offset	5.8%	6.7%	6.3%

Notes: Rows apply alternative equivalence scales or resource-offset assumptions to both the actual and counterfactual denominators; each wave is reanchored to its PIP headcount under the alternative scale. The Betson scale is $(A + 0.5K)^{0.7}$ for two or more adults; offset rows scale household resources down by 5 or 10 percent for each removed child.

Source: Author’s calculations from national household surveys anchored to the World Bank Poverty and Inequality Platform (PIP).

For every alternative scale, the actual and transported household denominators use the same formula, and every wave is reanchored to its PIP headcount (Table A2). The Betson scale is $(A + 0.5K)^{0.7}$ for two or more adults and $(1.3 + 0.5K)^{0.7}$ for one adult. Resource offset scenarios multiply resources by $(1 - o)^{K - K^{cf}}$ and are symmetric in forward and reverse calculations. Counting dependent children as ages 0–14 rather than 0–17, with the same pairs and count-weighted pooling, yields 11.7% for all persons, 17.7% for children, and 15.2% for people in households with children.

Measured child income is far smaller than either offset scenario. In the ENIGH and PNADC files, which record income by source for each household member, all members under 18 combined receive 3.5% of household income among Mexican households with children in 2016 (3.1% in 2024) and 1.6% among Brazilian households with children in 2023. Including childless households lowers these shares to 2.0% and 0.7%. A 5 percent offset for each removed child therefore assumes a larger income response than the combined measured contribution of all resident children. Children in these settings consume far more than they earn, and the fixed-resource baseline sits closer to the measured data than either scenario.

C Restricted identification set

The main paper reports the finite union of four economically motivated completions of latent ranks within observed child-count mass points. Random completion imposes no additional within-tie ordering. The copula completion calibrates dependence to the observed normal-score correlation of welfare and fertility. Age completion orders latent ranks by mother age. The monotone-selection completion assigns poorer mothers weakly higher latent fertility propensity, consistent with the sign of the observed cross-sectional gradient. The minimum and maximum over these declared completions form the reported restricted set. It is not the unrestricted set, and I do not claim sharpness under models outside these restrictions. Bounds over all couplings consistent with the observed marginals span zero, so some restriction on assignment is necessary for the data to be informative. The completions vary only the within-tie assignment; the rank-preservation condition itself is not varied. The reverse ordering, in which richer mothers have higher latent fertility within a tie, is computed as a diagnostic but excluded from the set because the observed welfare-fertility gradient is negative in every sample country and wave (Figure 1).

D IPUMS International extension

The nearest-anchor estimate holds fixed the closest survey welfare distribution (Table A3). The distant-anchor column repeats the exercise using the survey wave farthest in time from the census; anchor waves may include surveys outside the 1990–2019 estimation window. Large differences are evidence that the implied historical poverty effect is sensitive to the borrowed welfare distribution. No forward/reverse average is formed because the census does not observe comparable welfare. Mother age was not retained in the census marginal files, so the census transports are not age-standardized.

E Additional data that are not pooled

The Philippine FIES series observes under-15 child bands but not mother pointers in the historical welfare files. Household-level under-15 transports are reported as a separate sensitivity series (Table A4) because pooling them with children-per-mother would change the estimand. Demographic-only IPUMS countries likewise enter the fertility panel but not the welfare decomposition. This separation uses all of the data while keeping incompatible estimands apart.

Table A3: Earliest reliable one-directional IPUMS extension by country

Country	Census	Anchor	Relation	Nearest (pp)	Distant (pp)
Brazil	1960	1981	pre	2.02	4.33
Burkina Faso	1985	2018	pre	4.13	5.33
China	1982	1988	pre	1.36	5.43
Egypt	1986	1999	pre	0.00	1.84
Ethiopia	1984	1999	pre	-2.68	2.47
Ghana	1984	1991	pre	1.97	5.33
Indonesia	1971	1993	pre	3.43	12.80
Malawi	1987	2004	pre	-2.55	0.45
Mali	1987	2018	pre	-8.35	-9.15
Mexico	1970	1984	pre	1.71	3.55
Peru	2007	2004	interior	-1.99	—
Rwanda	1991	2000	pre	0.61	7.83
Senegal	1988	2018	pre	-9.01	-6.64
Uganda	1991	2009	pre	-6.19	-1.30

Notes: Relation describes where the census year falls relative to the country’s survey chain: pre for census years before the first survey wave, post for years after the last, and interior for years inside the window. A positive entry means the census family-size distribution implies more poverty than the anchor survey’s own distribution, so the survey-window estimate understates the longer-horizon contribution. The table displays the earliest reliable census for each country. The replication CSV reports every reliable census and all three poverty lines. Distant-anchor estimates use the survey wave farthest in time from the census and are sensitivity checks, not estimates to be averaged; a dash marks censuses whose farthest wave lacks usable mother links.

Source: Author’s calculations from IPUMS International census microdata and national household surveys anchored to the World Bank Poverty and Inequality Platform (PIP).

Table A4: Household-level under-15 fallback not included in the mother-level pool

Country	Window	Observed decline (pp)	Effect (pp)	Share
Philippines	1990–2009	17.3	6.0	34.9%

Notes: This transport uses under-15 children per household because the historical FIES files do not contain mother pointers. It is reported to use the available welfare series but is not pooled with children-per-mother estimates.

Source: Author’s calculations from the Philippine Family Income and Expenditure Survey anchored to the World Bank Poverty and Inequality Platform (PIP).